

Baron consultants

Total Solution Partner
for Digital Transformation of the construction industry

 Baron consultants


Baron consultants



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www.baroncs.co.kr

BARON Consultants, Co., Ltd. is dedicated to helping clients achieve sustainable construction by providing solutions that lead the digital transformation of the construction industry, enabling efficient project management and optimize design and construction processes.

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A Total Solution Partner

Driving Digital Transformation in the Construction Industry

Leveraging our construction engineering expertise and IT expertise, we provide core technologies and customized solutions that enable digital transformation throughout the infrastructure lifecycle.

PART. 1

Designing the Future

Baron Consultants

- 2 The Values Pursued by Baron
- 4 Future Presented by Baron
- 6 Technology realized by Baron

Management Motto

Corporate Philosophy

With **Technology**, let's make
Human & **Nature** come together



Our Value

We realize sustainable growth by transforming the paradigm of the construction industry through specialized construction engineering technologies and tools, and by fostering people-centric talent.



技

Deep Foundational Knowledge in Construction and Civil Engineering

Possession of High Expertise Based on Fundamental Understanding and Experience Across All Construction Phases

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High-Level Software Technology (Digital Technology)

Effective Integration of Diverse Technological Tools (S/W, H/W) and Processes

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Engineers/Technicians with Expertise in Various Fields

Accumulation of Knowledge Fusing Engineering Know-how and Digital Knowledge

材

Developers with Strong Digitalization Competency and Extensive Development Experience

Developers Capable of Creating Diverse Solution Software

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Acceptable Conditions/ Circumstances for Society, Businesses, and Systems

Shifting from Goal-Setting Guidelines to Implementation through Case Studies and Effectiveness Verification

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Capacity and Will to Undertake Continuous and Long-Term Investment

Large-Scale Investment for Technology Development and Continuous Investment for Process Innovation

Action Plan

We innovate the essence of the construction industry through Super-Gap Technology, challenging spirit, trust, and collaboration, and contribute to the sustainable construction sector as an agile enterprise.



Toward the Digital Transformation

The Construction Industry's Journey to Digital Transformation

To achieve a sustainable construction industry, We are leading the digital transformation of infrastructure lifecycle technologies.

Redefine through the Digital Transformation

Digital transformation is revolutionizing the construction industry

Process Innovation

- Digital transformation of analog-based work
- Innovation in the process of using location-based 3D models
- User-centric solutions and software

Product Innovation

- Improving the quality of existing products that have been standardized due to copy and paste
- In addition to analog books, digital information materials are also added
- Improving work efficiency using solutions

With Our Solution, Making Visions Real

Experience innovation through solutions

We offer diverse information and an optimal total solution so that all users the owner, contractor, and engineer can easily access and utilize phase data and achieve practical benefits.

Easy to Understand



Intuitive Usability



High Quality



Low Cost



Super Short



Core Technologies Enabling Digital Transformation

Key technologies for realizing digital transformation

We provide solutions for design, construction, operation, and maintenance through specialized construction software and hardware based on advanced construction knowledge and expertise.

Customized Civil Engineering Software

Customized Construction Engineering

We support digital transformation across the entire life cycle of infrastructure

Customized construction specialized Software

Customized support for the production and management of BIM deliverables

- Package program
- Solution



Research and Planning

By building a reliable GIS-based model, we support accurate information models

- Current Status Natural Environment •
- Current Status Social Environment •
- Surveying Inherent/Essence •
- Terrain Model •
- Subsurface Layer Model •

Operation and Maintenance

Based on asset information, we support proactive infrastructure maintenance and management

- As-Built BIM Model •
- BIM-IoT Integration •
- Real-Time Integrated Information Management System •



Design

Based on efficient technology verification and optimization, we support the production and management of high-quality digital deliverables

- Analysis and Optimization
- Simulation
- Structural Analysis
- Design BIM Model
- Digital Deliverable Creation/Management

Construction

Through a real-time communication and collaboration environment based on an accurate information model, we support fast and optimized decision-making

- Construction BIM Model
- Simulation
- Integrated Construction Information Management
- Construction Project Management
- BIM-IoT Integration
- Real-Time Collaboration System

The Digital Transformation across the Infrastructure Lifecycle

Optimal Space for Digital Integration

Optimal Environment for Digital Collaboration

We provide the optimal BIG Room environment based on an engineering-grade signage system

High-performance H/W for collaboration Hardware

Supports real-time collaboration and efficient decision-making

- Engineering signage
- Communication system



Display

- Integrated Management of Construction Information •
- High-Resolution Large Display •
- Large-Scale Infrastructure Customized •

Hardware System

- High-Specification PC for Large Software •
- Large-Capacity Storage High-Performance Server •
- High-Performance Network Environment •



Smart Construction + Decision-Making

- Digital Meeting System** • Video Conference • Drone, Site Control (CCTV) • IoT Sensors

Establishment of Optimal Collaboration Space

- The components of the BIG Room are implemented in a customized manner according to the spatial conditions of the site

Easy, Fast and Low-cost Customized Civil Engineering Solution

To support the Digital Transformation across the entire lifecycle of infrastructure facilities, we provide a total solution based on specialized construction knowledge and know-how.

PART. 2

Realizing Value

Services provided

12	The Values Pursued by Baron
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Digital Transformation for the Construction Industry

Innovation Roadmap

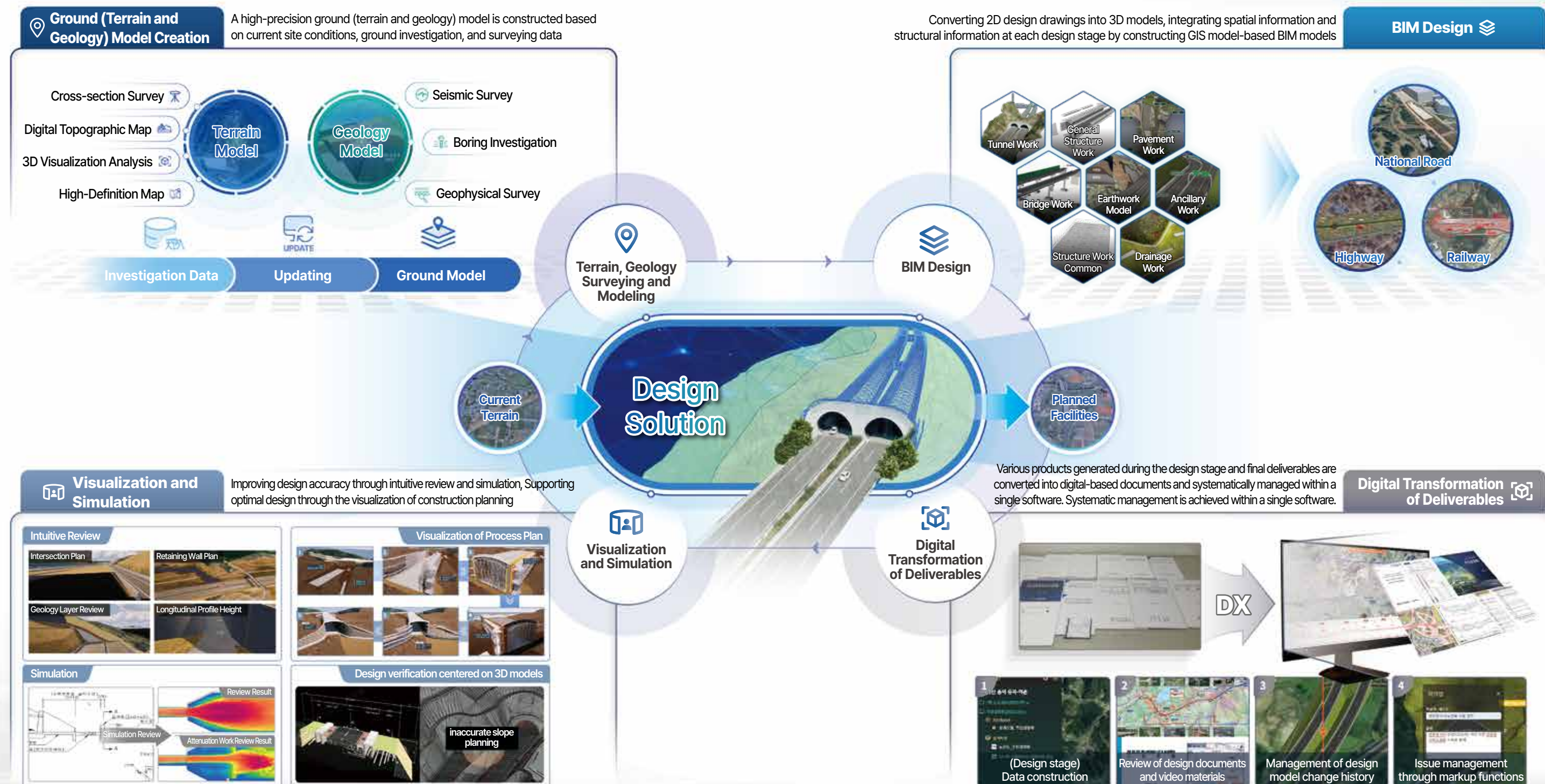
Specialized Software
for Expanding Infrastructure Construction Domains



Design Solution

By maximizing the efficiency, accuracy, and speed of design through the construction of high-precision ground models and GIS-based 3D BIM models, along with simulation and visualization, and by digitally transforming and integrally managing the deliverables generated during the design process, we realize optimal design quality and cost reduction.

We support the creation and integrated management of high-efficiency, ultra-high-quality digital BIM deliverables



Construction Solution

Based on a platform capable of integrally managing all construction information, we share and verify field data in real time, and maximize fast, accurate decision-making, process efficiency, and construction safety through VR/AR-based multi-party collaboration systems and immersive visuals.



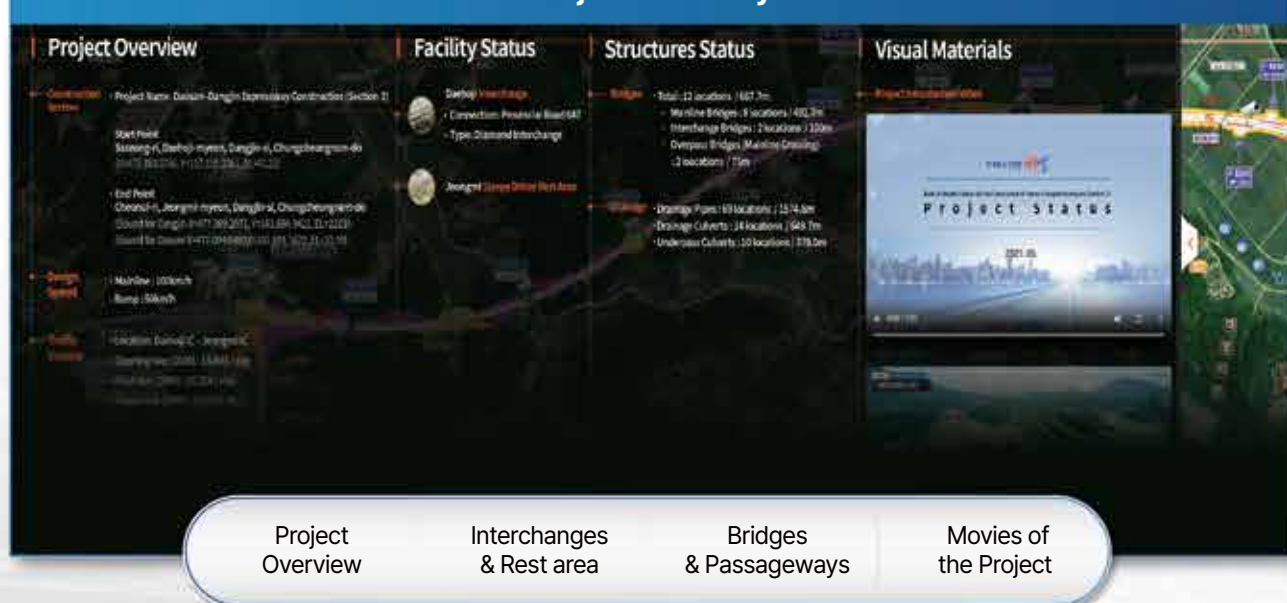
Project Information

The solution converts various products generated during the design stage (such as design reports and working drawings) and final deliverables (e.g., final submitted drawings) into digital-based documents, ensuring their integrated and systematic management within a single software platform.

Main Screen



Project Summary



Real-Time Collaboration for Accurate Decision-Making and Efficient Project Management

Construction Assistance



BIM 3D models are utilized to easily and intuitively review construction plans and design information.

Integrates and visualizes Geographic Information System (GIS) data with Road Facility Information (BIM)



Drone data—such as orthophotos, 3D mesh models, and point clouds—are overlaid with BIM models and drawings to compare construction progress

Measurement Function :
Point, Distance, Area, and Volume

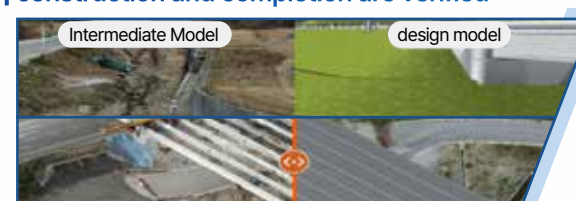
Immersive Video for a Realistic Experience

Immersive Video for a Realistic Experience Provide video footage that reflects seasonal, weather, and time-of-day conditions across various facilities

Facility Visualization :
Interchange, Bridge, Tunnel, Tollgate



Through comparative review between models, the consistency and status of construction and completion are verified



By comparing and reviewing the design model with the survey model generated via 3D digital surveying during construction, the consistency and status of the ongoing work are verified.

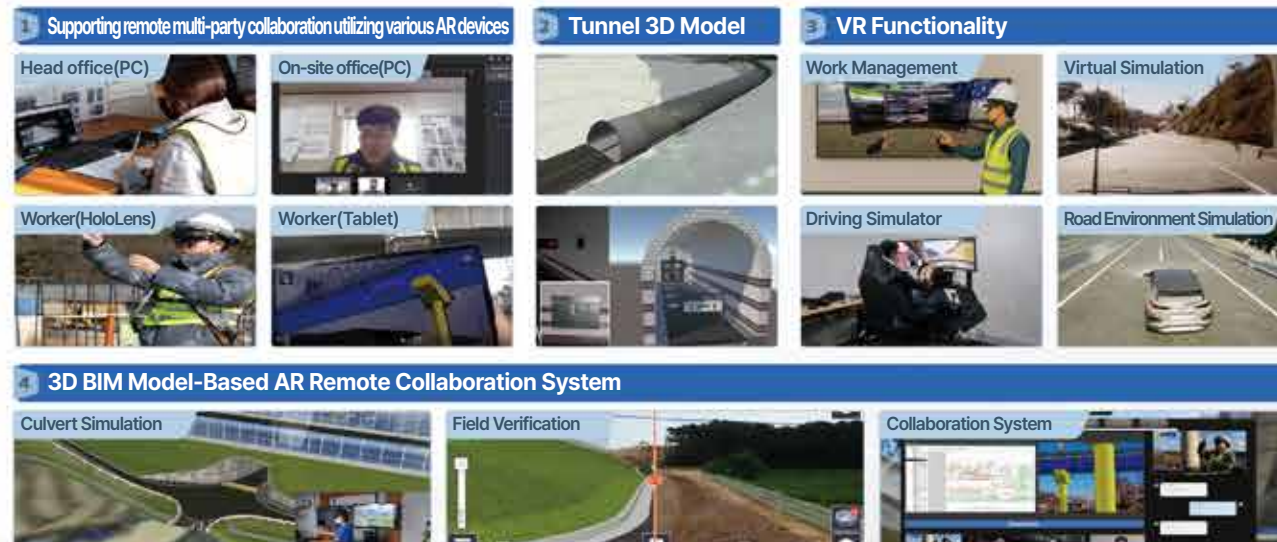


By integrating detailed construction models for each work type, verifies and confirms the completion status through a comparative review of the As-Built condition (measured via 3D surveying) and the final completion model.

Collaboration



Utilizing the design stage BIM model, the system integrates AR/VR collaboration features to support remote multi-party collaboration.



Operation & Maintenance Solution

By implementing a Digital Twin model of infrastructure facilities, collecting and monitoring the facility's status data, and providing various simulations, we maximize the safety of infrastructure and enhance the asset's value and lifespan in the long term through facility condition checks, proactive prediction of risks, and effective maintenance management.



Simulation

Utilizing Virtual Reality (VR) simulation, the system enhances the understanding of various physical and technical phenomena



Construction Simulation + Driving Simulation + Solar Simulation + Drainage Simulation

Real-time Operation & Maintenance



The solution implements facilities as a Digital Twin model and, based on this, collects and provides real-time status data of the facilities



Realistic Model for maintenance



Bridge Information



Damage to Piers



Damage to Pavement



Simulation



Real-time Operation and Maintenance



**Digital
Maintenance**

**Twin-Based
Solution**

Infrastructure Construction Expert Engineering Package software

Leveraging specialized knowledge across the entire lifecycle of infrastructure construction projects and IT expertise, we enhance operational efficiency and collaboration capabilities through dedicated software.

PART. 2

Realizing Value Services provided

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Beyond Analog, GAIA: Pioneering Digital Transformation in Geotechnical Investigation

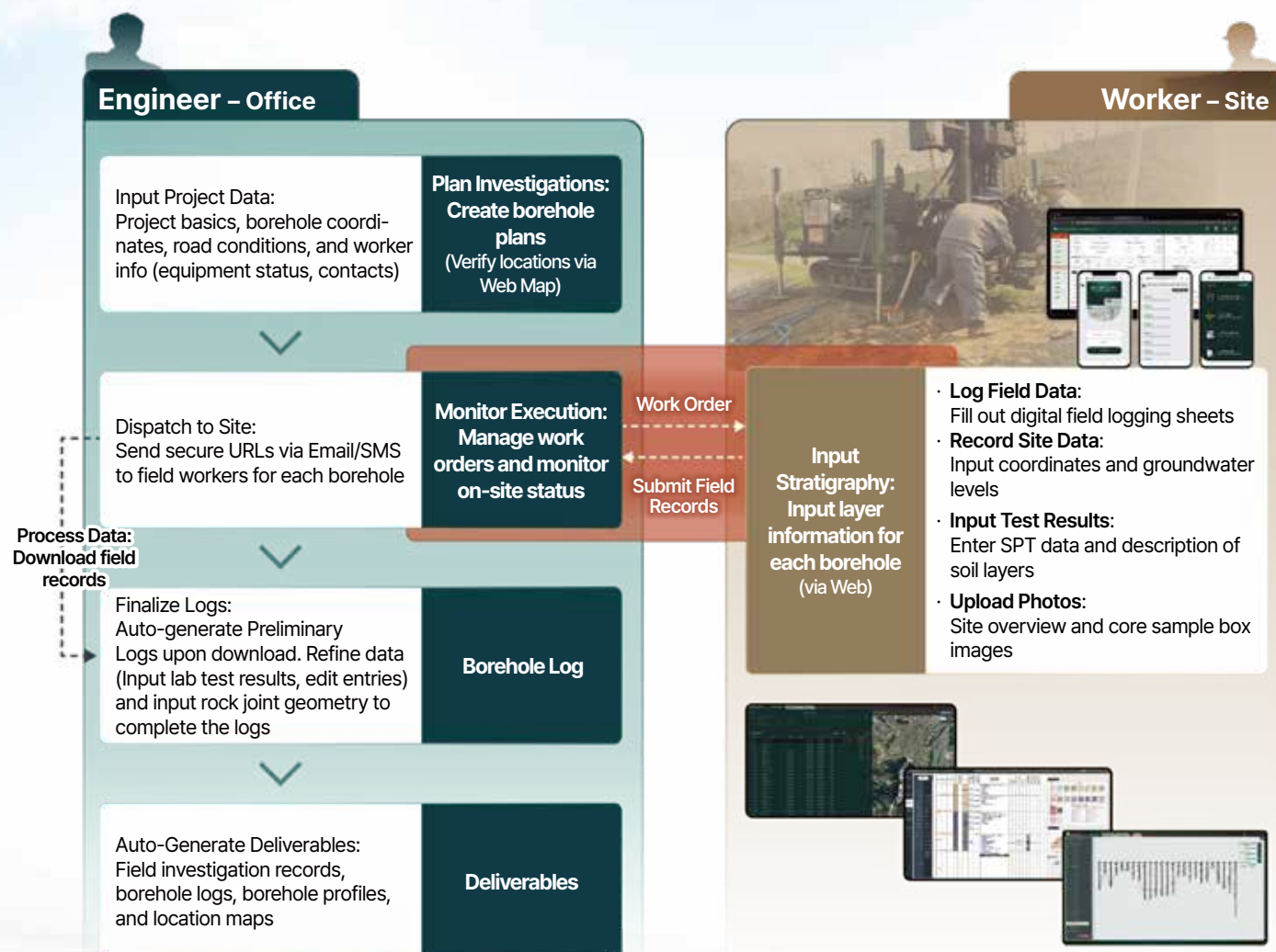
GAIA

Geotech Analytic Investigation Amass

www.k-gaia.co.kr



GAIA is a web-based service that digitizes geotechnical investigations and data management. From data collection to management and application, GAIA provides integrated management for all subsurface layer information.



Enter borehole investigation results in real time via mobile devices



Digital Transformation

Digitalize geotechnical investigation workflows by integrating IoT-based ground data, 3D & Geo BIM, and linkage with the National Ground Information Database



Enhancing Potential for Use as Smart Construction Technologies

Key Features

Web map-based borehole investigations and mobile field logs, with real-time stratigraphy input and automatic generation of borehole logs, enable reliable investigation management



Reliable data and efficient drilling investigations

Deliverables

Deliver high-quality outputs, including geotechnical location maps, borehole field records, and borehole log profiles



Diverse, user-friendly, and high-quality outputs

Korea's Premier Integrated Transportation Demand Analysis Program
TOVA leads the future of transportation

TOVA

Total Vehicle Assignment

www.k-tova.com



TOVA supports optimal transportation system planning by comprehensively analyzing and forecasting passenger demand for road, bus, urban rail, intercity rail, aviation, and maritime transport.

1



Integrated Passenger & Freight
Transportation Demand Analysis Software

2



Global UI Platform
Optimized for local languages and regional settings

3



Seamless Integration and Maximized
Utilization of KTDB Data

4



GIS-based
Transportation Network Analysis Environment

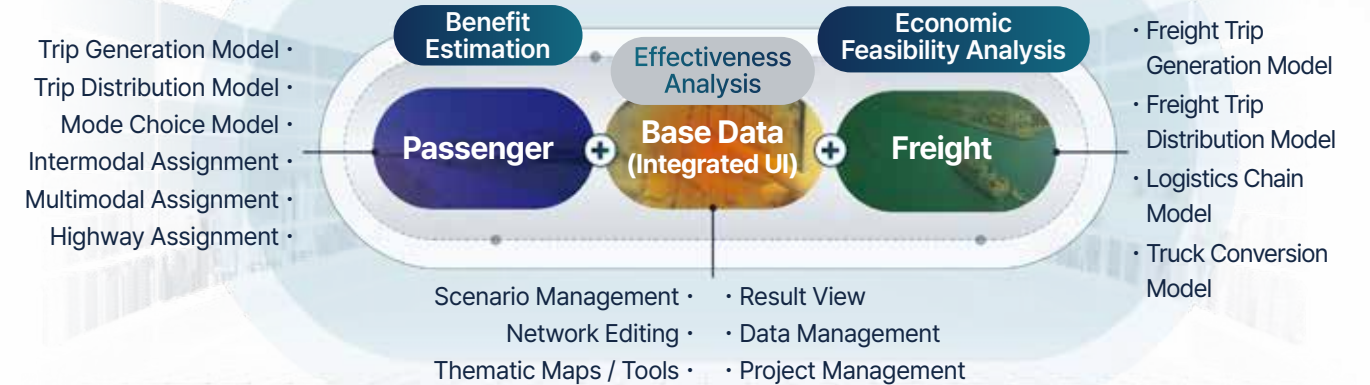
5



Comprehensive
Analysis Tools for Diverse
Transportation Policies



Keywords for System



Key Features

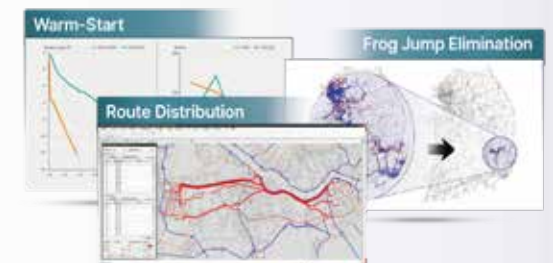
Provides advanced analysis features ranging from split-screen views and network comparison to traffic volume and public transit route analysis



The first traffic demand analysis platform enabling multimodal assignment

Road Network Analysis

Enhances the accuracy and reliability of road traffic assignment, ensuring balanced results across all routes and vehicle types



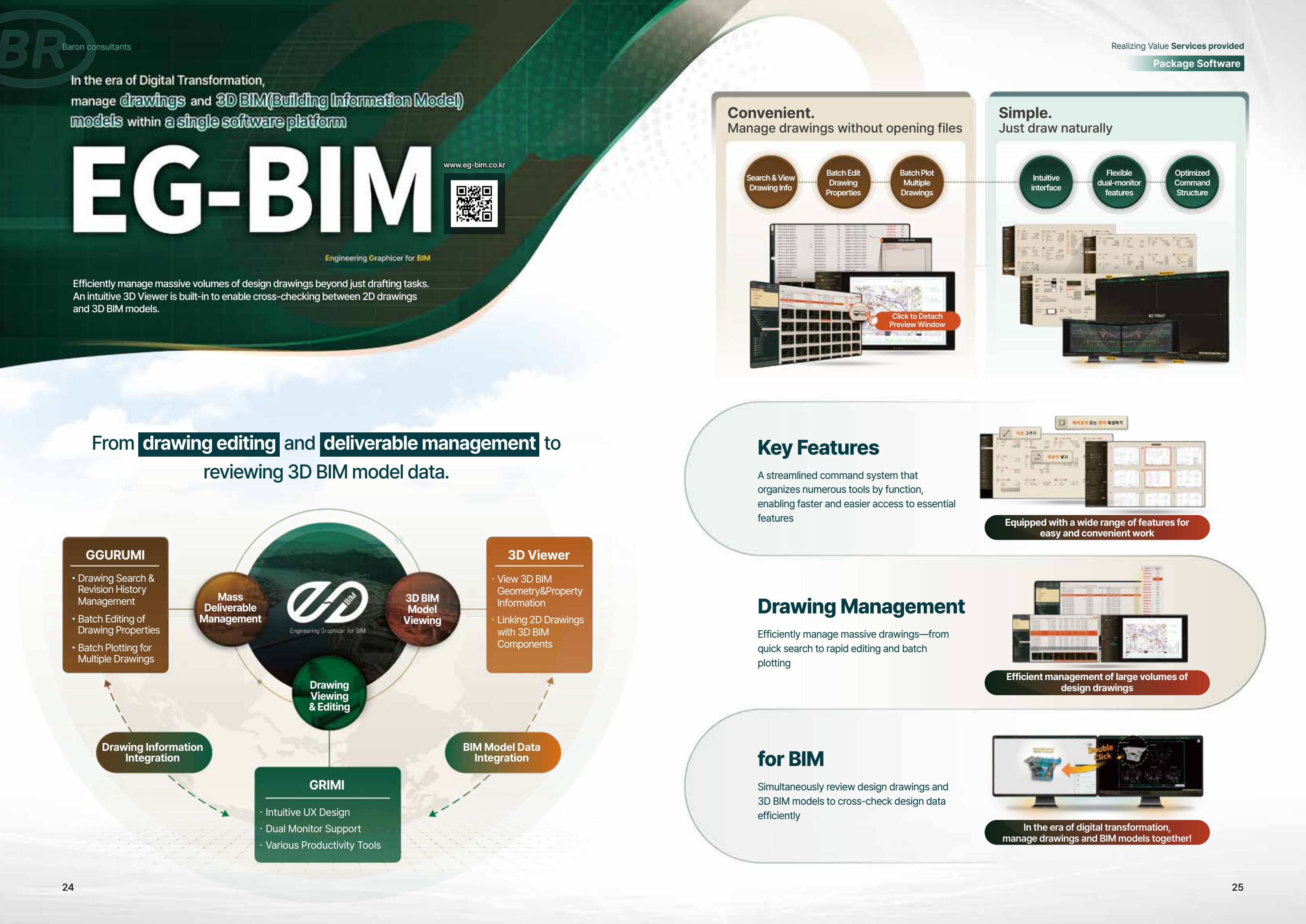
Demand and performance evaluation and analysis for road transportation systems

Public Transportation Analysis

Explores all realistically available public transit routes to improve the reliability and explanatory power of analysis results



Comprehensive route finding across all public transportation modes



In the era of Digital Transformation,
manage drawings and 3D BIM(Building Information Model)
models within a single software platform

EG-BIM

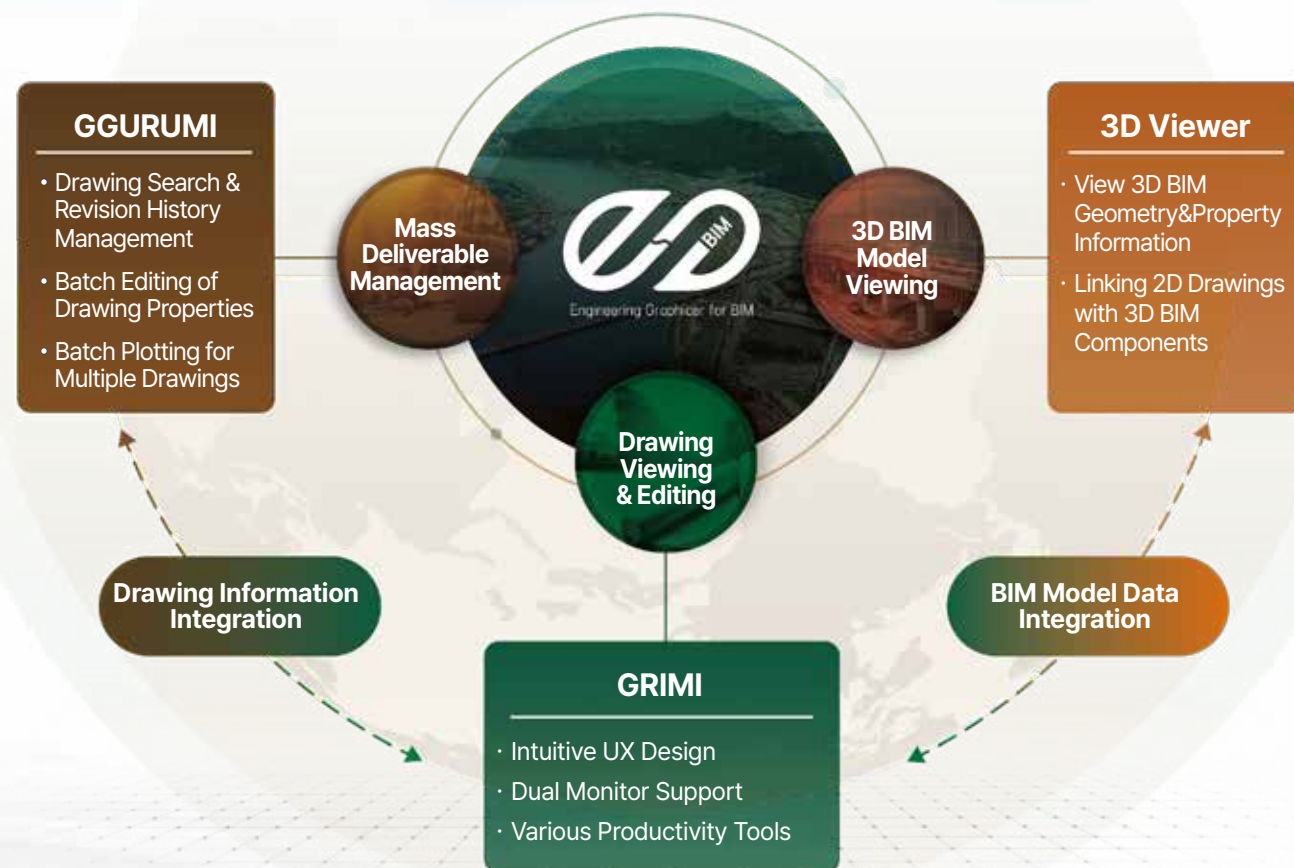
www.eg-bim.co.kr



Engineering Graphicer for BIM

Efficiently manage massive volumes of design drawings beyond just drafting tasks.
An intuitive 3D Viewer is built-in to enable cross-checking between 2D drawings
and 3D BIM models.

From **drawing editing** and **deliverable management** to
reviewing 3D BIM model data.



Convenient.

Manage drawings without opening files



Simple.

Just draw naturally



Key Features

A streamlined command system that organizes numerous tools by function, enabling faster and easier access to essential features



Equipped with a wide range of features for easy and convenient work

Drawing Management

Efficiently manage massive drawings—from quick search to rapid editing and batch plotting



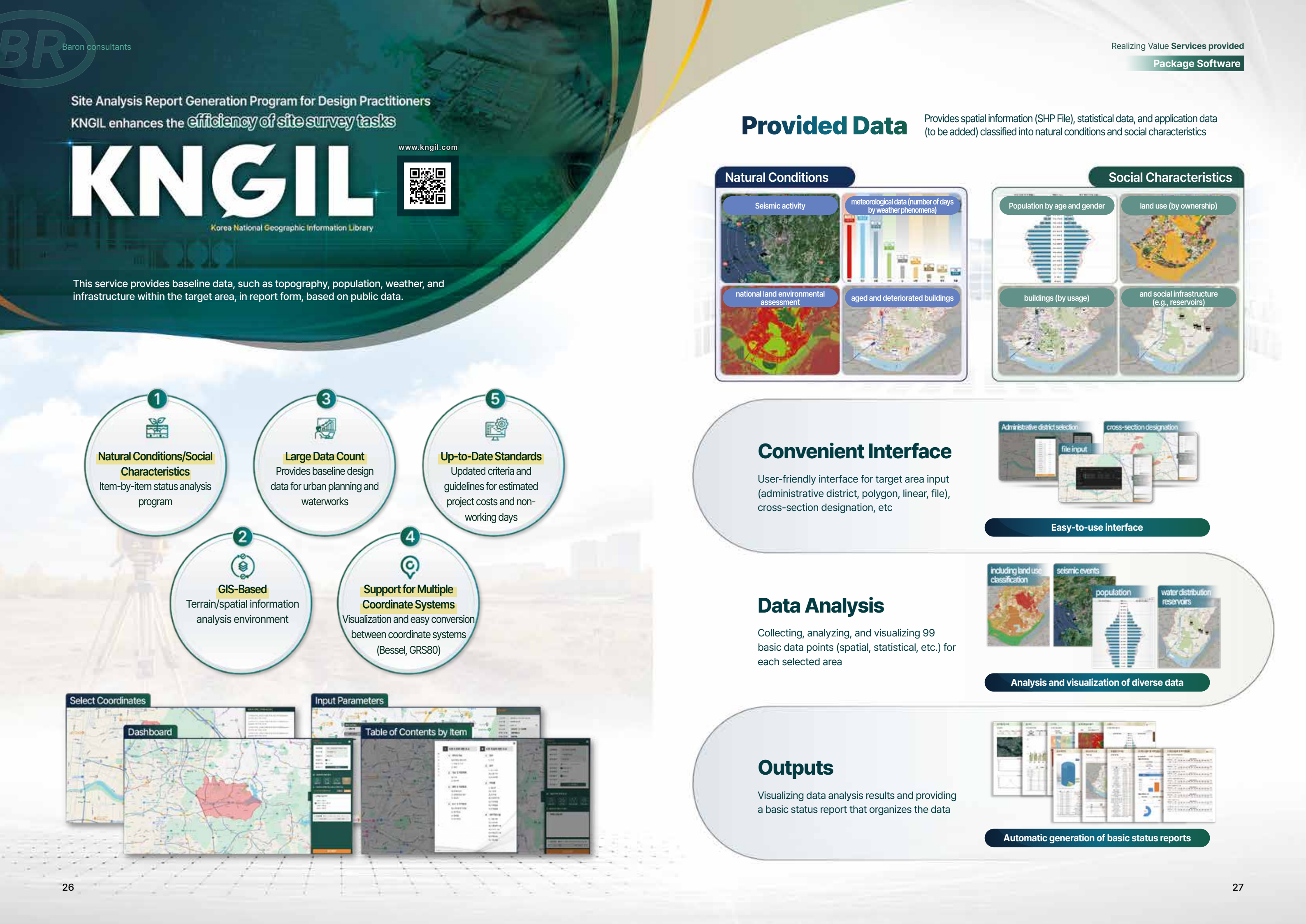
Efficient management of large volumes of design drawings

for BIM

Simultaneously review design drawings and 3D BIM models to cross-check design data efficiently



In the era of digital transformation, manage drawings and BIM models together!



Site Analysis Report Generation Program for Design Practitioners
KNGIL enhances the efficiency of site survey tasks

KNGIL

Korea National Geographic Information Library

www.kngil.com



This service provides baseline data, such as topography, population, weather, and infrastructure within the target area, in report form, based on public data.

1



Natural Conditions/Social Characteristics

Item-by-item status analysis program

3



Large Data Count

Provides baseline design data for urban planning and waterworks

5



Up-to-Date Standards

Updated criteria and guidelines for estimated project costs and non-working days

2



GIS-Based

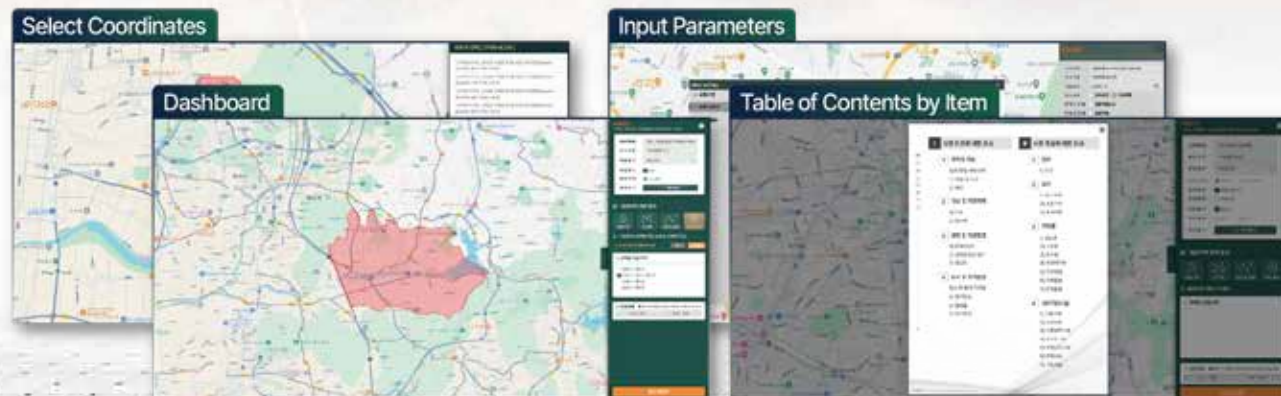
Terrain/spatial information analysis environment

4



Support for Multiple Coordinate Systems

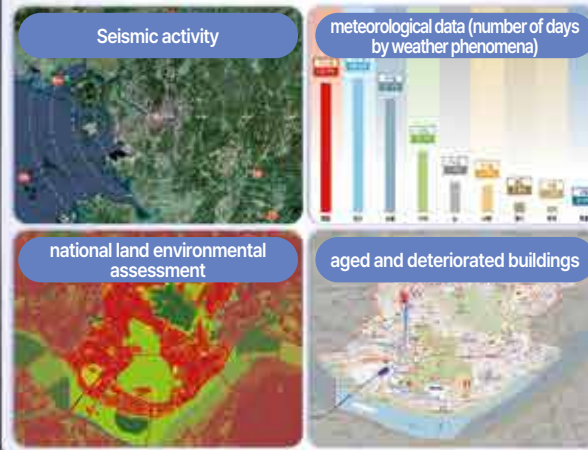
Visualization and easy conversion between coordinate systems (Bessel, GRS80)



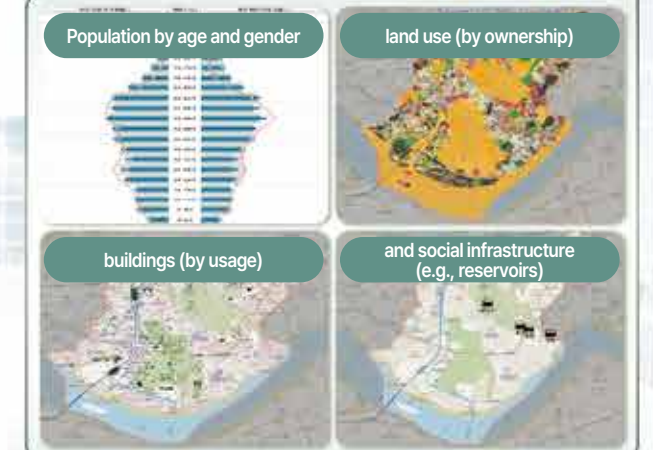
Provided Data

Provides spatial information (SHP File), statistical data, and application data (to be added) classified into natural conditions and social characteristics

Natural Conditions



Social Characteristics



Convenient Interface

User-friendly interface for target area input (administrative district, polygon, linear, file), cross-section designation, etc



Easy-to-use interface

Data Analysis

Collecting, analyzing, and visualizing 99 basic data points (spatial, statistical, etc.) for each selected area



Analysis and visualization of diverse data

Outputs

Visualizing data analysis results and providing a basic status report that organizes the data



Automatic generation of basic status reports

An Integrated Sewer Network Design Solution from Hydraulic Analysis to Automated Network Design

iPipes

Intelligent Pipe Network Design & Analysis Sewer

This specialized sewer network design program supports longitudinal and cross-sectional design of sewer pipes and provides drawing and water quantity calculation report generation functions.

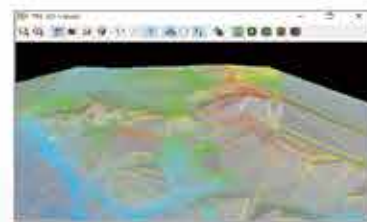
Input of existing pipe network data

- ☑ Pipe network construction using existing data such as FLO, INP, SHP, and KML.



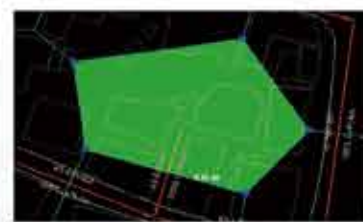
Automatic generation of terrain information

- ☑ Generation of triangulated network through recognition of elevation points and contour lines from digital topographic maps
- ☑ Stratigraphic interpolation based on borehole data.



Automatic input of catchment area

- ☑ Automatic division of watershed areas based on pipes and manholes and automatic input of catchment areas.



Integrated plan design

- ☑ Input and editing of the status data of pipe networks, watersheds, land use zones, roads, complexes, and underground facilities.



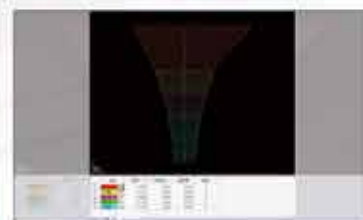
Reflection of road/complex zoning information

- ☑ Drawing of left and right cross-section boundaries using road/complex zoning lines.



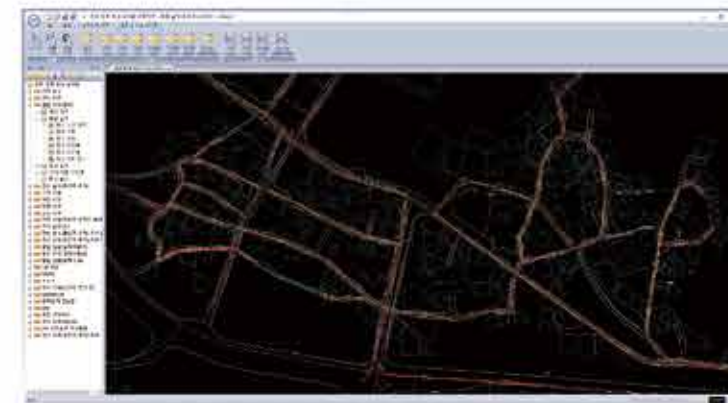
Design of various cross-slopes

- ☑ Setting of slope styles for each section, displaying of ground layers and groundwater levels
- ☑ Distinction between on-land and underwater excavations.



Convenient Pipe Network Construction

Convenient pipe network data construction through pipe network input using existing data and input of current data such as watershed boundaries, roads/complex divisions, boreholes, and underground utilities



Automatic Pipe Network Design and Analysis

- Extraction of ground elevations at pipeline start and end points using TIN
- Obstacle collision avoidance during automatic design
- Customized design using flow velocity simulation
- Reporting of analysis results (flow velocity, cover depth, slope, and overflow)



Automatic Input of Longitudinal Parameters

- Configuration of pavement, construction, temporary works and foundation types using the longitudinal parameter input function
- Automatic input of temporary works section information by setting temporary works application criteria based on each excavation depth (H)
- Configuration of pipe foundation types based on pipeline code (pipe diameter/type)



Generation of Design Deliverables

- Creation of hydraulic calculation reports and quantity reports by work type in Excel format, and production of drawing deliverables (pipe network diagram, longitudinal diagram, cross-section diagram, cross-section development diagram, etc.) through EG-BIM integration



Creating a New Standard for PQ Work
through Data Integration and Document Automation

PQ Program

Pre-Qualification Program

From bid announcement management to efficient management of various data, including technicians and performance, we support the entire PQ project process.

“Based on a database, it enables easy management of the performance of **construction engineering contractors and technicians** and provides **automated** PQ document creation”

Information Collection and Management

Systematic management of all data required for PQ, including schedules, personnel, and performance

Document Creation and Review

Production of high-quality PQ submission documents through automated documentation and customized review processes

Bid information management

Technician information management

Company performance management

Efficient management, reliable data, and high-quality outcomes

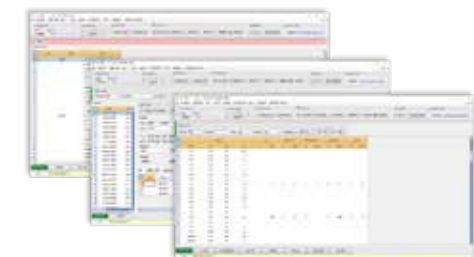
Support for PQ document creation

Industry-specific customized review

Output of deliverables

Key Features

Integrated management of diverse data, including bidding information, engineers, and company performance. Customized review features tailored to industry requirements. Digitalization of the PQ preparation process to maximize efficiency



All-in-one management of PQ tasks, from bidding to documentation

Project Management

Manage projects based on company performance, link engineers and data to provide a comprehensive at-a-glance overview, and provide efficient operational support, including attachment management



Manage performance, engineers, and the overall project at a glance

Deliverables

PQ documents generated automatically, tailored to the client's required format, compatible with Excel and HWP, and required attachments organized to deliver high-quality final submission documents



Standardized PQ documents generated automatically to provide complete support through submission

Customized Hardware System for the **BIG Room**

All-in-One Real-Time Collaboration and Decision Support

PART. 2

Realizing Value

Services provided

12	The Values Pursued by Baron
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BIG Room

We offer an optimal project collaboration hardware package that includes conference systems, engineering signage, and dedicated servers, for issue review and rapid decision-making



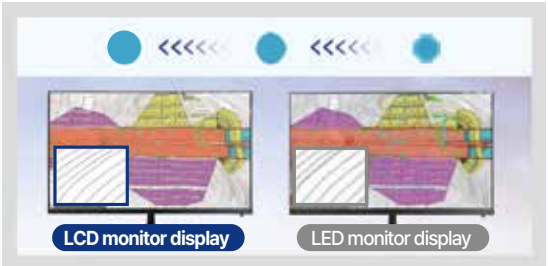
BIG Room for the Infrastructure Construction



Precision

Precision rendering of complex linear data and 3D models displaying vector-based drawings and 3D model information with exceptional clarity and accuracy

BigRoom: The Hub for Real-Time Collaboration



Infrastructure Customized

For large infrastructure projects such as roads, railways, and bridges, engineering signage enables comprehensive visualization and review of vast amounts of information at a glance



Direct execution and real-time performance

A simple and efficient system architecture allows the latest information to be executed directly without delay, ensuring real-time situational awareness and supporting immediate, informed decision-making

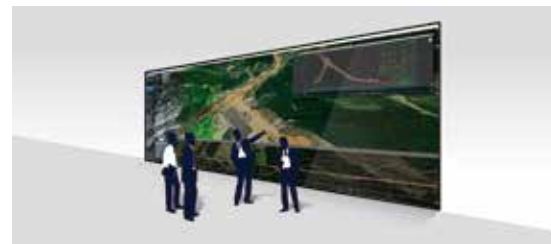




Hardware Package Configuration

High-Resolution Large Display

Utilizing engineering-optimized large-format displays, we create an integrated environment that enables intuitive understanding of construction data through precise analysis and clear visualization of design results



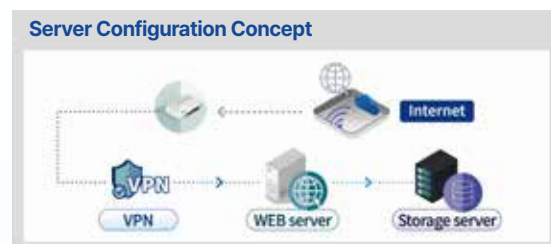
High-Performance PCs for Large-Scale Infrastructure

High-specification PCs provide a stable environment for software handling complex terrain and structural geometries

Recommended specifications for smooth operation	Minimum specifications defined to ensure cost efficiency
✓ i9 CPU ✓ 128GB RAM ✓ RTX4090	✓ i7 CPU ✓ 64GB RAM ✓ RTX4070 TI

High-Performance Servers for BIM Data

High-performance storage and servers for large amounts of data, including extensive spatial information and drone footage, ensure a stable BIM environment



for Collaboration Digital Conference System

Real-time collaboration is possible anytime, anywhere, without spatial constraints, and a digital conferencing system supports immediate decision-making from design to construction



Implementation Cases



Implementation of XR-based Smart Construction BIG Room System
Hamyang-Hapcheon Construction Project Unit
Korea Expressway Corporation
(Hanwha Corporation E&C Division), 2023



Smart Construction BIG Room Implementation and Operation Support
Seosan-Asan Construction Project Unit
Korea Expressway Corporation
(HANSHIN Engineering Construction), 2024



Smart Construction BIG Room Implementation and Operation Support
Daesan-Dangjin Sections 1 and 2
Korea Expressway Corporation
(KOLON GLOBAL, KCC E&C), 2024



Smart Construction BIG Room Implementation and Operation Support
Daesan-Dangjin Sections 3 and 4
Korea Expressway Corporation (KUMHO E&C), 2024